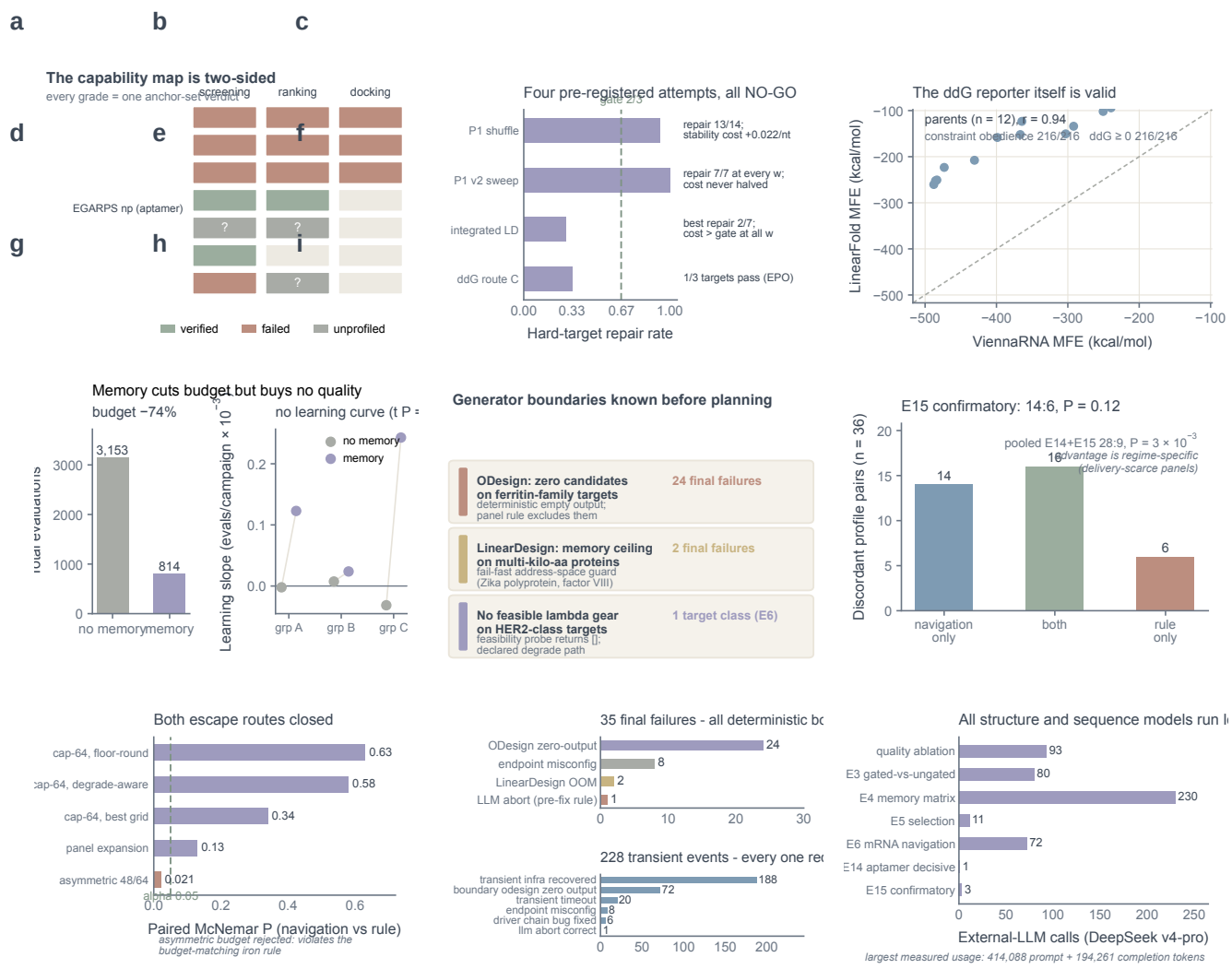

a
b
c
The capability map is two-sided
every grade = one anchor-set verdict
screening
ranking
docking
d
e
f
EGARPS np (aptamer)
g
h
i
verified
failed
unprofiled
Four pre-registered attempts, all NO-GO
gate 2/3
P1 shuffle
repair 13/14;
stability cost +0.022/nt
P1 v2 sweep
repair 7/7 at every w;
cost never halved
integrated LD
best repair 2/7;
cost > gate at all w
ddG route C
1/3 targets pass (EPO)
0.00
0.33
0.67
1.00
Hard-target repair rate
The ddG reporter itself is valid
parents (n = 12) r = 0.94
constraint obedience 216/216
ddG ≥ 0 216/216
LinearFold MFE (kcal/mol)
ViennaRNA MFE (kcal/mol)
Memory cuts budget but buys no quality
budget −74%
no learning curve (t P =
3,153
814
no memory
memory
Total evaluations
Learning slope (evals/campaign × 10
grp A
grp B
grp C
Generator boundaries known before planning
ODesign: zero candidates on ferritin-family targets
deterministic empty output; panel rule excludes them
24 final failures
LinearDesign: memory ceiling on multi-kilo-aa proteins
fail-fast address-space guard (Zika polyprotein, factor VIII)
2 final failures
No feasible lambda gear on HER2-class targets
feasibility probe returns []; declared degrade path
1 target class (E6)
E15 confirmatory: 14:6, P = 0.12
pooled E14+E15 28:9, P = 3 × 10
advantage is regime-specific
(delivery-scarce panels)
Discordant profile pairs (n = 36)
navigation only
both
rule only
Both escape routes closed
cap-64, floor-round
cap-64, degrade-aware
cap-64, best grid
panel expansion
asymmetric 48/64
alpha 0.05
Paired McNemar P (navigation vs rule)
asymmetric budget rejected: violates the budget-matching iron rule
35 final failures - all deterministic bc
ODesign zero-output
endpoint misconfig
LinearDesign OOM
LLM abort (pre-fix rule)
228 transient events - every one rec
transient infra recovered
boundary odesign zero output
transient timeout
endpoint misconfig
driver chain bug fixed
llm abort correct
All structure and sequence models run l
quality ablation
E3 gated-vs-ungated
E4 memory matrix
E5 selection
E6 mRNA navigation
E14 aptamer decisive
E15 confirmatory
External-LLM calls (DeepSeek v4-pro)
largest measured usage: 414,088 prompt + 194,261 completion tokens